



Vectors

Key terms

vector, magnitude, scalar, linear combination, dot product

KSBs: K13, S11, B1, S53

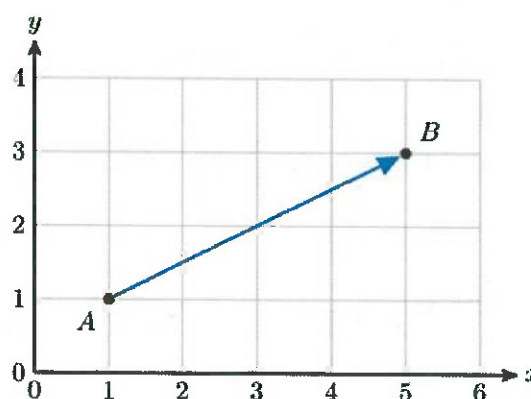
8.1 Vectors

Vectors are a way of expressing quantities that cannot be described by a *single number*. This could involve a set of data, or a single point in a data set with *multiple variables*. It can also be interpreted geometrically.

Vectors are sometimes thought of as being an 'arrow' in n -dimensional space. We can describe the location of a point in space using *coordinates for x and y* (or x , y and z in 3D). A vector describes the *difference between two points* as an arrow from one to the other.

For example, consider a vector in 2D space: there is a distance in one direction, and a distance in another, so a *single value* cannot carry this information.

Vectors were originally used in geometry, but have since become important in data science, computer graphics, engineering and physics.



Definition 8.1 - Vector

A **vector** is a collection of numbers, called *entries* written as an *ordered list in a bracket*, which defines a *change of position in n -dimensional space*.

We usually denote vectors by *single lower-case letters* like $\mathbf{v}$ or $\mathbf{a}$, and the entries in the vectors with subscript numbers, written as a row or column:

$$\underline{\mathbf{a}} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = (a_1, a_2, \dots, a_n)$$

To make it clear what we're talking about, the character representing a vector is written in boldface when typeset, e.g., $\mathbf{a}$ or underlined when written by hand, e.g., $\mathbf{a}$.

Exercise 8.1. In the statement below, underline x whenever it refers to the vector.

"The vector x is defined as $x = (x_1, x_2, x_3)$. The x -coordinate of x is x_1 ."

We use $\mathbf{0}$ to denote the zero vector $(0, 0, \dots, 0)$. This can be thought of as the vector $a - a$, or the vector from a point a to itself.

8.1.1 Vector Magnitude

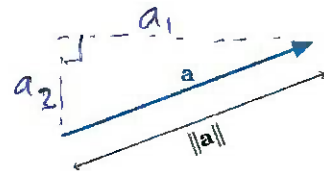
The **magnitude** of a vector $\mathbf{a}$ is the straight-line distance between the head and tail of $\mathbf{a}$. In two dimensions, this can be calculated using Pythagoras' Theorem and in higher dimensions we can use an extended version of it.

Definition 8.2 - Vector Magnitude

In 2D, the magnitude of a vector $\mathbf{a} = (a_1, a_2)$, denoted by $\|\mathbf{a}\|$, is calculated as $\sqrt{a_1^2 + a_2^2}$.

For a general vector $\mathbf{a} = (a_1, a_2, \dots, a_n)$, $\|\mathbf{a}\|$ is given by

$$\|\mathbf{a}\| = \sqrt{\sum a_i^2} = \sqrt{a_1^2 + a_2^2 + \dots + a_n^2}$$



For any two vectors $\mathbf{a}, \mathbf{b}$ and any scalar k , the following properties are satisfied:

- $\|\mathbf{a}\| \geq 0$ (the magnitude of a vector is always positive)
- $\|\mathbf{a}\| = 0 \iff \mathbf{a} = \mathbf{0}$ (the magnitude is zero if and only if the vector is the zero vector).

Example 8.1. Let $\mathbf{v} = (2, 7, 1)$. Then the magnitude of $\mathbf{v}$ can be calculated as:

$$\|\mathbf{v}\| = \sqrt{2^2 + 7^2 + 1^2} = \sqrt{4 + 49 + 1} = \sqrt{54} = 7.34.$$

Exercise 8.2. Calculate the magnitudes of the following vectors:

- $\mathbf{p} = (3, 4)$
- $\mathbf{u} = (1, 2, 3)$
- $\mathbf{v} = (5, -12, 0)$
- $\mathbf{w} = (1, 0, 1)$

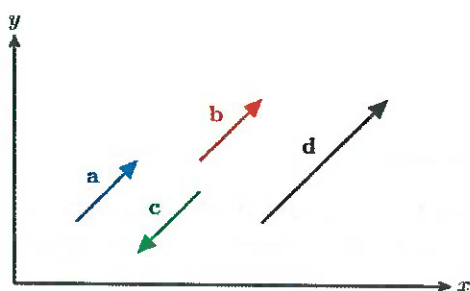
8.1.2 Arithmetic operations on vectors

Definition 8.3 - Vector Equality

Two vectors are said to be equal if *their corresponding entries are equal*. In other words, $\mathbf{a} = (a_1, a_2, \dots, a_n)$ and $\mathbf{b} = (b_1, b_2, \dots, b_n)$ respectively, are equal if and only if the elements which correspond co-ordinate-wise are equal, i.e.,

$$a_1 = b_1, a_2 = b_2, \dots, a_i = b_i \quad i = 1, \dots, n.$$

Informally, this happens if and only if two vectors *have the same magnitude* and *point in the same direction*.



Here, the vectors $\mathbf{a}$ and $\mathbf{b}$ point in the same direction and have the same magnitude, so we can say that

$$\mathbf{a} = \mathbf{b}.$$

The vector $\mathbf{c}$ has *the same magnitude as $\mathbf{a}$ and $\mathbf{b}$* but points in *the opposite direction*, so $\mathbf{a} \neq \mathbf{c}$. The vector $\mathbf{d}$ points in the same direction as $\mathbf{a}$ but has a larger magnitude - so $\mathbf{a} \neq \mathbf{d}$.

Definition 8.4 - Addition of two vectors

The sum of two vectors $\mathbf{a}$ and $\mathbf{b}$ can be calculated as

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{pmatrix}$$

A geometric representation of the sum $\mathbf{a} + \mathbf{b}$ is shown in the figure.

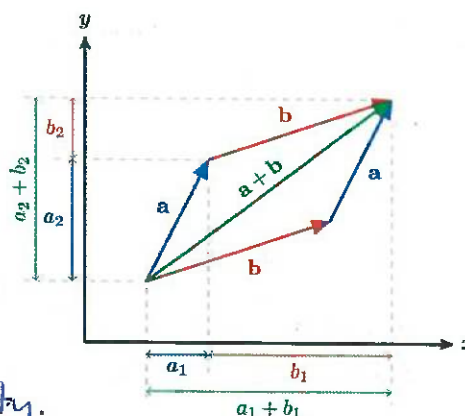
We place the vector $\mathbf{a}$, and then position the vector $\mathbf{b}$ so *its tail is at the head of $\mathbf{a}$* . The endpoint of $\mathbf{b}$ is *the head of the vector $\mathbf{a} + \mathbf{b}$* .

Note that the order in which we add the two vectors is *unimportant* - if we placed $\mathbf{b}$ first, then $\mathbf{a}$, we would get to the same point. This is known as the *parallelogram law of addition*.

For two vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|.$$

This property is called the ~~triangle~~ triangle inequality.



Example 8.2. Let $\mathbf{a} = (1, 0, 2)$ and $\mathbf{b} = (2, -3, 1)$. Then

$$\mathbf{a} + \mathbf{b} = (1, 0, 2) + (2, -3, 1) = (1+2, 0+(-3), 2+1) = (3, -3, 3)$$

Exercise 8.3. Let $\mathbf{u} = (2, 3)$, $\mathbf{v} = (3, -2)$ and $\mathbf{w} = (1, 6)$. Write down

- (a) $\mathbf{u} + \mathbf{v}$
- (b) $\mathbf{u} + \mathbf{w}$
- (c) $\mathbf{v} - \mathbf{w}$

8.1.3 Scalar multiplication

Definition 8.5 - Scalar

A scalar is a single numerical value, which can be a whole number, decimal or fraction.

We can multiply a vector by a scalar to get another vector.

Definition 8.6 - Scalar Multiplication

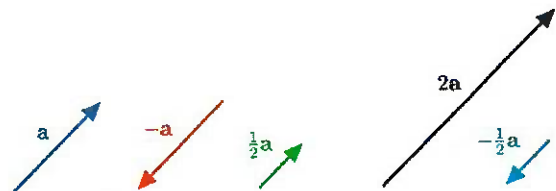
The product of a vector $\mathbf{a}$ by a scalar k is given by

$$k\mathbf{a} = k \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} ka_1 \\ ka_2 \\ \vdots \\ ka_n \end{pmatrix}$$

Multiplying a vector by a scalar has the effect of scaling the vector $\mathbf{a}$ to produce a vector parallel to $\mathbf{a}$ with a magnitude of $k \cdot \|\mathbf{a}\|$.

The product of a vector $\mathbf{a}$ and a scalar k has the following properties:

- $k\mathbf{a}$ is a vector parallel to $\mathbf{a}$, i.e. $\mathbf{a} \parallel k\mathbf{a}$
- $0\mathbf{a} = \mathbf{0}$
- $1\mathbf{a} = \mathbf{a}$
- $\|k\mathbf{a}\| = |k| \|\mathbf{a}\|$
- If $k > 0$ then $\mathbf{a}$ and $k\mathbf{a}$ point in the same direction, whereas if $k < 0$ then $\mathbf{a}$ and $k\mathbf{a}$ point in opposite directions



Example 8.3. Let $\mathbf{v} = (1, 2, 3, 4)$ and $k = 6$. Then $k\mathbf{v} = 6 \times (1, 2, 3, 4) = (6 \times 1, 6 \times 2, 6 \times 3, 6 \times 4) = (6, 12, 18, 24)$.

Exercise 8.4. Calculate $k\mathbf{v}$, where $\mathbf{v} = (2, 3, 4)$ and

- (a) $k = 2$
- (b) $k = 1.5$
- (c) $k = 1$
- (d) $k = 0$

8.1.4 Linear combination of vectors

Definition 8.7 - Linear combination of vectors

Let $\mathbf{v}, \mathbf{u}_1, \dots, \mathbf{u}_m$ be n -dimensional vectors such that:

$$\mathbf{v} = \alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2 + \dots + \alpha_m \mathbf{u}_m$$

for scalars $\alpha_1, \alpha_2, \dots, \alpha_m$. Such a sum is called a *linear combination of vectors*.

Note

α is the lowercase Greek character *alpha* and is equivalent to 'a' in the Latin alphabet.

For example:

$$\begin{pmatrix} 2 \\ 0 \\ 7 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix} + \begin{pmatrix} 0 \\ -10 \\ 9 \end{pmatrix}$$

Here we have expressed $(2, 0, 7)$ as a linear combination of the vectors $(1, 5, -1)$ and $(0, -10, 9)$. Later we will see how this can help us to solve systems of related linear equations.

8.1.5 Dot Product

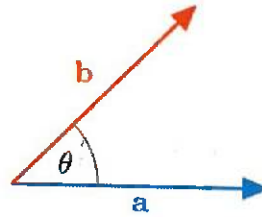
Definition 8.8 - The dot product

The **dot product** (also known as the **scalar product**) of two vectors $\mathbf{a}$ and $\mathbf{b}$ is defined as

$$\mathbf{a} \cdot \mathbf{b} = \sum a_i b_i = a_1 b_1 + a_2 b_2 + \dots + a_n b_n.$$

The dot product of two vectors $\mathbf{a}$ and $\mathbf{b}$ can be defined in geometric terms as

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cdot \cos(\theta)$$



where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.

Roughly, we can think of the dot product $\mathbf{a} \cdot \mathbf{b}$ as a scalar that describes the amount of agreement between $\mathbf{a}$ and $\mathbf{b}$. If two vectors point in roughly the same direction we would expect their dot product to be larger than the dot product of two vectors pointing in completely different directions.

From the definition of the dot product, we can say:

- The dot product produces a scalar quantity (a number)
- The dot product of two perpendicular vectors is zero, and we write $\mathbf{a} \perp \mathbf{b}$ if $\mathbf{a} \cdot \mathbf{b} = 0$.
- The dot product of two parallel vectors $\mathbf{a}$ and $\mathbf{b}$ is $\|\mathbf{a}\| \|\mathbf{b}\|$.
- $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$.
- $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2 \geq 0$ and if $\mathbf{a} \cdot \mathbf{a} = 0$ then we must have $\mathbf{a} = \mathbf{0}$.

Example 8.4. Let $\mathbf{a} = (1, 2, 3)$ and $\mathbf{b} = (3, -1, 0)$. The dot product of $\mathbf{a}$ and $\mathbf{b}$ is:

$$\mathbf{a} \cdot \mathbf{b} = (1 \times 3) + (2 \times -1) + (3 \times 0) = 3 - 2 + 0 = 1$$

Exercise 8.5. Which pairs of the following vectors are perpendicular?

$$\mathbf{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \mathbf{v} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} \quad \mathbf{w} = \begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$

Function: SUMPRODUCT (E/S)

Recall that the `=SUMPRODUCT()` function will take as inputs two ranges of the same size, and will return the sum of the pairwise products. This is the same calculation as the dot product.

8.2 Vectors in Data Science

The dot product is a useful tool to combine two vectors, and has several uses in data science contexts.

Example 8.5. Recall that data can be presented in a frequency table.

To find the mean of the data in the table, we can multiply each value by its frequency and add them together to get the value of $\sum x$, then divide this by the total frequency.

If we stored these two sets of data as vectors, we would have

$$v = (1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9) \quad f = (1 \ 0 \ 1 \ 3 \ 2 \ 0 \ 1 \ 1 \ 2)$$

Then

$$v \cdot f = (1 \times 1) + (2 \times 0) + \dots + (9 \times 2) = 59$$

This can be divided by the sum of the frequencies to get the mean of the dataset, $\frac{59}{11} = 5.36$.

Value	Frequency
1	1
2	0
3	1
4	3
5	2
6	0
7	1
8	1
9	2

8.2.1 Covariance

Recall that the covariance of two random variables X and Y is given by

$$\text{cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n-1}$$

Theorem 8.1 - Covariance and dot product

Let X and Y be two random variables, both with a mean of zero. Then

$$\text{cov}(X, Y) = \frac{X \cdot Y}{n-1}$$

That is, the dot product of the two variables (considered as vectors), divided by $n-1$.

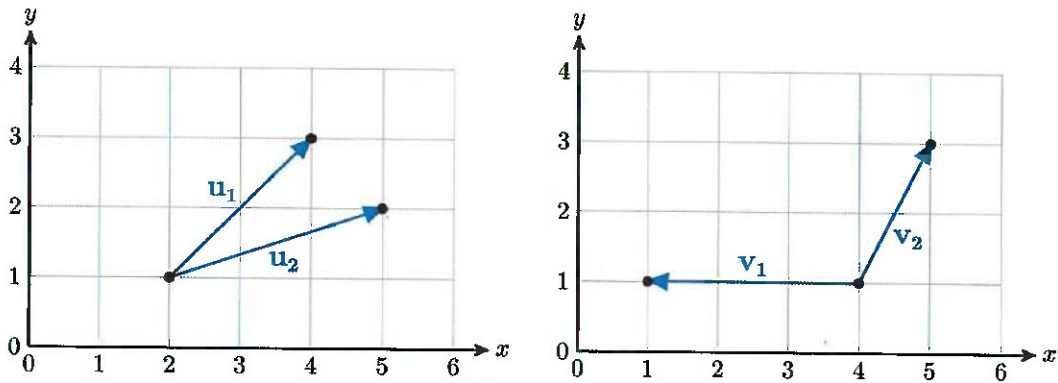
Indeed, if the means of both datasets are zero, we have

$$\text{cov}(X, Y) = \frac{\sum (x-0)(y-0)}{n-1} = \frac{\sum (x_i \times y_i)}{n-1}$$

This is just the definition of the dot product on top of the fraction.

8.2.2 Vector Agreement

Recall that the dot product of two vectors can be thought of as a measure of how much the two vectors *are pointing in the same direction*.



In the image, u_1 and u_2 are pointing in *roughly the same direction*, while v_1 and v_2 are pointing in *very different directions*. Recall that if two vectors are perpendicular (at right angles), their dot product is *zero*.

We would expect $u_1 \cdot u_2$ to be *positive* and $v_1 \cdot v_2$ to be *negative*.

$$\underline{u}_1 \cdot \underline{u}_2 = (2, 2) \cdot (3, 1) = 8 \quad \underline{v}_1 \cdot \underline{v}_2 = (-3, 0) \cdot (1, 2) = -3$$

If we have data points in n dimensions representing n variables of linked data, if two vectors representing data points are *in agreement* they are likely to have *similar values* and this can allow us to categorise how similar two observations are.

This can be applied in any number of dimensions, and is used in machine learning, image recognition and many other applications.